# WASP-50b

**R-0782**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_221121` · created 2026-07-28T06:19:51+00:00

**BENCHMARK: NON-DETECTION**

## Results

| | |
|---|---|
| **Mid-Transit Time (Tmid)** | 2459904.755 +/- 0.072 BJD_TDB |
| **Ratio of Planet to Stellar Radius (Rp/R*)** | 0.168 +/- 0.047 |
| **Transit depth (Rp/Rs)^2** | 2.83 +/- 1.57 [%] |
| **Orbital Inclination (inc)** | 82.45 +/- 2.73 |
| **Airmass coefficient 1 (a1)** | 1.263 +/- 0.014 |
| **Airmass coefficient 2 (a2)** | -0.165 +/- 0.027 |
| **Scatter in the residuals of the lightcurve fit is** | 1.11 % |
| **Best Comparison Star** | None |
| **Optimal Aperture** | 4.88 |
| **Optimal Annulus** | 6.76 |
| **Transit Duration (day)** | 0.031 +/- 0.031 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| **Rp/R■ fit vs published** | 0.168 ± 0.047 vs 0.1404 (deviation 0.37σ) |
| **Duration fit vs published** | 0.031 d vs 0.0754 d (deviation 0.9σ) |
| **Mid-time O–C** | -42.0 min |
| **Depth SNR** | 2.3 |
| **Residual scatter** | 1.11 % |

## Light curve

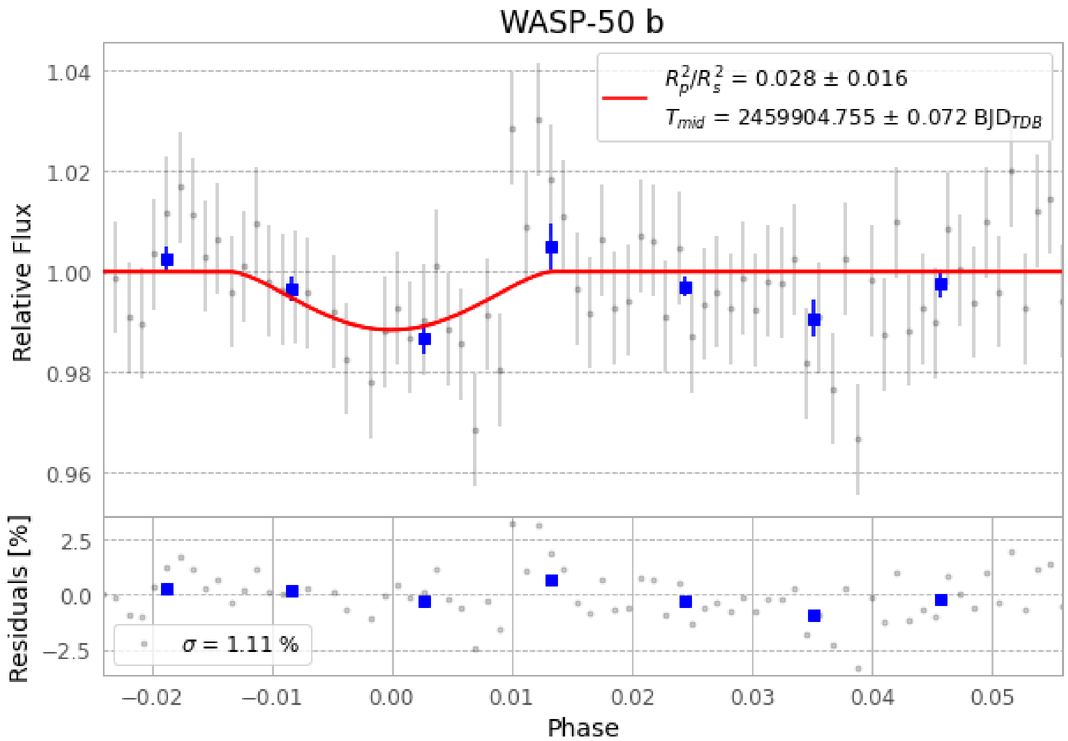


Figure 1 — FinalLightCurve_WASP-50 b_2022-11-20.png (SHA-256 851641825b7c5f02…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 334], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c9eeaf5a85216b67…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

## Data provenance

76 FITS frames (50 MB), `WASP-50221121045115.FITS` → `WASP-50221121083615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-221121.log (919a81dc7229…), FinalLightCurve_WASP-50 b_2022-11-20.png (851641825b7c…), FinalLightCurve_WASP-50 b_2022-11-20.csv (37032492c9d7…)

## Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)